

SALON & SPA APPOINTMENT BOOKING SYSTEM

1. Introduction

The Salon & Spa Appointment Booking System is a Java-based desktop application developed to manage salon and spa appointments efficiently. The system allows users to enter customer details, select a stylist and service, choose a date and time, and manage appointments through a graphical user interface. The application uses Java Swing for the GUI and object-oriented programming concepts to organize the different parts of the system.

2. Objectives

- To provide an easy-to-use salon appointment booking system.
- To store customer and stylist information.
- To manage different salon services and their prices.
- To book, search, update and cancel appointments.
- To prevent a stylist from being booked for the same date and time.
- To display appointments in an organized table.
- To demonstrate Java OOP concepts and GUI programming.

3. Technologies Used

- **Programming Language:** Java
- **GUI Framework:** Java Swing
- **IDE:** IntelliJ IDEA
- **Collections:** ArrayList, List and HashMap
- **Date and Time:** LocalDate, LocalTime and LocalDateTime
- **Concepts:** OOP, Interfaces, Exception Handling, Generics, Lambda Expressions and Event Handling

4. Project Structure

Main.java: Starts the application and launches the GUI.

SalonGUI.java: Provides the graphical user interface and handles user interaction.

Customer.java: Stores customer details such as name and phone number.

Stylist.java: Stores stylist details such as name and specialty.

ServicePackage.java: Stores the service name and price.

ServiceCategory.java: Defines fixed service categories using an enum.

Appointment.java: Represents a complete appointment containing customer, stylist, service, date and time.

AppointmentManager.java: Handles the main appointment management and business logic.

Bookable.java: Interface defining booking operations.

BookingException.java: Custom exception class for booking-related errors.

5. System Flow

The basic flow of the application is:

Main.java → SalonGUI → AppointmentManager → Appointment → Customer / Stylist / ServicePackage

Main.java starts the application and opens SalonGUI. The GUI collects information from the user and creates the required objects. The appointment is then passed to AppointmentManager, which performs operations such as booking, searching, updating and cancelling. The results are displayed through the GUI.

6. Customer Class

The Customer class represents a customer in the salon system. It contains two private fields: name and phone. A constructor is used to initialize these values, while getters and setters provide controlled access to them. The class demonstrates encapsulation by keeping its data private.

7. Stylist Class

The Stylist class represents a stylist working at the salon. It stores the stylist's name and specialty. The constructor initializes these values, and getters and setters are used to access and modify them. The stylists in the project include Rahul, Priya, Amit and Neha, each having a different specialty.

8. ServicePackage Class

The ServicePackage class represents a service provided by the salon. It stores the service name and price. The constructor initializes the service details, while getters and setters allow the values to be accessed and modified.

The services used in the project are:

- Premium Haircut – ₹500
- Hair Spa – ₹800
- Luxury Facial – ₹1200

- Full Body Massage – ₹1800

9. ServiceCategory Enum

ServiceCategory is an enum used to define a fixed set of service categories. The categories in the project are HAIR, SKIN and WELLNESS. An enum is useful when the possible values are fixed and predefined.

10. Appointment Class

The Appointment class represents one complete appointment. It contains the appointment ID, customer, stylist, service, date and time. It connects objects from the Customer, Stylist and ServicePackage classes into one appointment. The class uses constructors, getters, setters, encapsulation and method overriding through the toString() method.

11. Bookable Interface

Bookable is an interface that defines the main appointment operations of the system. It declares methods for booking, updating and cancelling appointments. AppointmentManager implements this interface and provides the actual implementation of these methods.

12. AppointmentManager Class

AppointmentManager is the main business logic class of the project. It manages appointments and performs operations such as booking, searching, updating, cancelling and sorting appointments.

It uses HashMap to store appointments using their IDs. Another HashMap called timeMap is used to prevent the same stylist from

being booked at the same date and time. An ArrayList is used to maintain service history.

The class also provides searching by customer and stylist, sorting by price and date/time, appointment ID generation and service history management.

13. Appointment Booking

When the user books an appointment, the GUI collects the customer's name, phone number, stylist, service, date and time. The system creates the required objects and creates an Appointment object. The appointment is then passed to AppointmentManager.

Before booking, the system checks whether the selected stylist is already booked for the selected date and time. If the time slot is already occupied, the appointment is rejected.

14. Search, Update and Cancel

The system allows appointments to be searched using their appointment ID. Existing appointments can also be updated with new customer, stylist, service, date or time information. Appointments can be cancelled using their ID, after which the appointment is removed from the appointment records and its time slot becomes available.

15. SalonGUI

SalonGUI is the graphical user interface of the project and extends JFrame. It uses Java Swing components such as JLabel, JTextField, JComboBox, JButton and JTable.

The GUI allows the user to enter information and interact with the appointment system. It uses ActionListener and lambda expressions to handle events such as button clicks and stylist selection.

When a stylist is selected, the GUI automatically displays their specialty and selects the corresponding service. For example, selecting Amit displays Massage & Wellness and selects Full Body Massage.

16. Java Swing Components Used

JFrame: Provides the main application window.

JLabel: Displays text and information.

TextField: Allows the user to enter information.

ComboBox: Provides dropdown selections for stylists and services.

Button: Provides buttons for actions such as booking, searching, updating and cancelling.

JTable: Displays appointment records in rows and columns.

17. Object-Oriented Programming Concepts

Encapsulation: Data is kept private inside classes and accessed through getters and setters.

Inheritance: SalonGUI extends JFrame and BookingException extends Exception, demonstrating inheritance.

Abstraction: The Bookable interface hides implementation details and defines the required appointment operations.

Polymorphism: AppointmentManager provides implementations of the methods defined by the Bookable interface.

Constructors: Constructors are used to initialize objects such as Customer, Stylist, ServicePackage and Appointment.

Method Overriding: The toString() method is overridden to provide meaningful text representations of objects.

18. Collections Used

HashMap: Used to store appointments using a key-value structure. The appointment ID is used as the key and the Appointment object is the value.

ArrayList: Used for storing service history and temporary lists of appointments.

List: Used when methods need to return or work with collections of appointments.

Generics: Used to specify the type of data stored in collections, such as Map<Integer, Appointment> and List<Appointment>.

19. Exception Handling

The project includes a custom BookingException class that extends Exception and is designed for booking-related errors. The current AppointmentManager implementation uses RuntimeException for situations such as an unavailable stylist time slot or an appointment not being found. The project also uses try-catch for handling exceptions during GUI initialization.

20. Date and Time Handling

The project uses Java date and time classes. LocalDate represents a date, LocalTime represents a time, and LocalDateTime represents both date and time. DateTimeFormatter is used to format and parse date and time values.

21. Main Features

- Customer information management
- Stylist selection
- Service selection
- Automatic stylist specialty display
- Automatic service selection
- Appointment booking
- Appointment searching
- Appointment updating
- Appointment cancellation
- Duplicate booking prevention
- Search by customer
- Search by stylist
- Sorting by price
- Sorting by date and time
- Service history
- Appointment table display

22. Advantages

The system provides a simple and user-friendly way to manage salon appointments. It reduces manual appointment management, prevents duplicate stylist bookings, organizes appointment information and demonstrates the practical use of Java object-oriented programming and GUI development.

23. Future Scope

The project can be further improved by adding a database for permanent data storage, user login and authentication, online payment functionality, email or SMS notifications, stylist availability management and an online/web-based version of the application.

24. Conclusion

The Salon & Spa Appointment Booking System demonstrates how Java can be used to develop a complete desktop application using object-oriented programming and Java Swing. The project combines classes, objects, encapsulation, inheritance, interfaces, collections, exception handling, event handling and GUI components to provide an appointment management system. The application successfully manages customers, stylists, services and appointments through an interactive graphical interface.